NOTESDB : 2 (DDL AND DML COMMANDS)

THEORY

1. Introduction

A Notes Management System is designed to help users create, organize, and manage notes efficiently. It allows users to categorize notes, set reminders, and attach files. The system is based on a relational database structure to ensure data consistency and easy retrieval.

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2. Entities and Attributes

- **User**

Represents the person using the system.
Attributes:
User_ID (Primary Key)
First_Name, Middle_Name, Last_Name
Email (Unique)
Password
Age

- **Category**

Used to organize notes into groups.
Attributes:
Category_ID (Primary Key)
User_ID (Foreign Key)

- **Note**

Stores the main content created by users.
Attributes:
Note_ID (Primary Key)
Title
Description
Priority
Category_ID (Foreign Key)
User_ID (Foreign Key)
Status
Created_At, Updated_At

- **Reminder**

Helps users remember important notes.
Attributes:
Reminder_ID (Primary Key)
Note_ID (Foreign Key)
Reminder_Time
Is_Complete

- **Attachment**

Stores files linked to notes.
Attributes:
Attachment_ID (Primary Key)
Note_ID (Foreign Key)

File_Path
File_Type

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3. Relationships

- **User – Creates – Note**

One user can create many notes.
Relationship: 1 : N

- **User – Organizes – Category**

One user can create multiple categories.
Relationship: 1 : N

- **Category – Contains – Note**

One category can have many notes.
Relationship: 1 : N

- **Note – Has – Reminder**

One note can have multiple reminders.
Relationship: 1 : N

- **Note – Has – Attachment**

One note can have multiple attachments.
Relationship: 1 : N

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4. ER Model Explanation

The ER (Entity-Relationship) model represents the logical structure of the database.
Entities are represented as rectangles.
Attributes are shown as ovals.
Relationships are represented using diamonds.
Primary keys are underlined.
Foreign keys establish connections between tables.
This design ensures:
Minimal data redundancy
Proper data integrity
Easy data retrieval

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5. Constraints Used

Primary Key: Ensures unique identification (e.g., User_ID, Note_ID)
Foreign Key: Maintains relationships between tables
Unique Constraint: Email must be unique
Check Constraint: Age must be between 1 and 100

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6. Advantages of This Design

Organized data structure
Easy to manage and scale
Supports multiple users
Efficient searching and filtering
Maintains data consistency

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7. Applications

Personal note-taking apps

Task management systems
Reminder applications

QUERIES:

Study and project planners

```
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 18
Server version: 8.0.45 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| airplanedb         |
| company            |
| company_db         |
| companydb          |
| emp                |
| employee           |
| employee_appraisal |
| employee_db        |
| information_schema |
| mysql              |
| notesdb            |
| performance_schema |
| sakila             |
| smart_notekeeper   |
| sql_detective      |
| sys                |
| world              |
+--------------------+
17 rows in set (0.01 sec)

mysql> USE notesdb;
Database changed

mysql> CREATE TABLE User (
    ->    User_ID INT PRIMARY KEY,
    ->    First_Name VARCHAR(50),
    ->    Middle_Name VARCHAR(50),
    ->    Last_Name VARCHAR(50),
    ->    Email VARCHAR(100) UNIQUE,
```

```
   ->    Password VARCHAR(50),
   ->    Age INT CHECK (Age BETWEEN 1 AND 100)
   -> );
Query OK, 0 rows affected (0.08 sec)

mysql> CREATE TABLE Category (
   ->    Category_ID INT PRIMARY KEY,
   ->    User_ID INT,
   ->    FOREIGN KEY (User_ID) REFERENCES User(User_ID)
   -> );
Query OK, 0 rows affected (0.05 sec)

mysql> CREATE TABLE Note (
   ->    Note_ID INT PRIMARY KEY,
   ->    Title VARCHAR(100),
   ->    Description TEXT,
   ->    Priority VARCHAR(20),
   ->    Category_ID INT,
   ->    User_ID INT,
   ->    Status VARCHAR(20),
   ->    Created_At DATETIME,
   ->    Updated_At DATETIME,
   ->    FOREIGN KEY (Category_ID) REFERENCES Category(Category_ID),
   ->    FOREIGN KEY (User_ID) REFERENCES User(User_ID)
   -> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE Reminder (
   ->    Reminder_ID INT PRIMARY KEY,
   ->    Note_ID INT,
   ->    Reminder_Time DATETIME,
   ->    Is_Complete BOOLEAN,
   ->    FOREIGN KEY (Note_ID) REFERENCES Note(Note_ID)
   -> );
Query OK, 0 rows affected (0.09 sec)

mysql> CREATE TABLE Attachment (
   ->    Attachment_ID INT PRIMARY KEY,
   ->    Note_ID INT,
   ->    File_Path VARCHAR(200),
   ->    File_Type VARCHAR(50),
   ->    FOREIGN KEY (Note_ID) REFERENCES Note(Note_ID)
   -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO User VALUES
   -> (1,'Payal','A','Charvande','payal@gmail.com','pass123',21),
   -> (2,'Rahul','B','Sharma','rahul@gmail.com','rahul123',23);
Query OK, 2 rows affected (0.02 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> INSERT INTO Category VALUES
```

```
   -> (101,1),
   -> (102,2);
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> INSERT INTO Note VALUES
   -> (201,'DBMS','Study ER model','High',101,1,'Active',NOW(),NOW()),
   -> (202,'CN','Study DHCP','Medium',102,2,'Pending',NOW(),NOW());
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> INSERT INTO Reminder VALUES
   -> (301,201,'2026-04-10 10:00:00',0),
   -> (302,202,'2026-04-11 12:00:00',0);
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> ^C
mysql> INSERT INTO Attachment VALUES
   -> (401,201,'/files/dbms.pdf','PDF'),
   -> (402,202,'/files/cn.docx','DOCX');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM User;
+---------+------------+-------------+-----------+-----------------+----------+------+
| User_ID | First_Name | Middle_Name | Last_Name | Email           | Password | Age  |
+---------+------------+-------------+-----------+-----------------+----------+------+
|       1 | Payal      | A           | Charvande | payal@gmail.com | pass123  |   21 |
|       2 | Rahul      | B           | Sharma    | rahul@gmail.com | rahul123 |   23 |
+---------+------------+-------------+-----------+-----------------+----------+------+
2 rows in set (0.00 sec)

mysql> SELECT * FROM Category;
+-------------+---------+
| Category_ID | User_ID |
+-------------+---------+
|         101 |       1 |
|         102 |       2 |
+-------------+---------+
2 rows in set (0.00 sec)

mysql> SELECT * FROM Note;
+---------+-------+----------------+----------+-------------+---------+---------+---------------------+---------------------+
| Note_ID | Title | Description    | Priority | Category_ID | User_ID | Status  | Created_At          | Updated_At          |
+---------+-------+----------------+----------+-------------+---------+---------+---------------------+---------------------+
|     201 | DBMS  | Study ER model | High     |         101 |       1 | Active  | 2026-04-08 21:53:39 | 2026-04-08 21:53:39 |
|     202 | CN    | Study DHCP     | Medium   |         102 |       2 | Pending | 2026-04-08 21:53:39 | 2026-04-08 21:53:39 |
+---------+-------+----------------+----------+-------------+---------+---------+---------------------+---------------------+
2 rows in set (0.00 sec)

mysql> SELECT * FROM Reminder;
+-------------+---------+---------------------+-------------+
```

```
| Reminder_ID | Note_ID | Reminder_Time | Is_Complete |
+-------------+---------+---------------------+-------------+
|         301 |     201 | 2026-04-10 10:00:00 |           0 |
|         302 |     202 | 2026-04-11 12:00:00 |           0 |
+-------------+---------+---------------------+-------------+
2 rows in set (0.00 sec)

mysql> SELECT * FROM Attachment;
+---------------+---------+-----------------+-----------+
| Attachment_ID | Note_ID | File_Path       | File_Type |
+---------------+---------+-----------------+-----------+
|           401 |     201 | /files/dbms.pdf | PDF       |
|           402 |     202 | /files/cn.docx  | DOCX      |
+---------------+---------+-----------------+-----------+
2 rows in set (0.00 sec)

mysql> SELECT Title, Priority FROM Note;
+-------+----------+
| Title | Priority |
+-------+----------+
| DBMS  | High     |
| CN    | Medium   |
+-------+----------+
2 rows in set (0.00 sec)

mysql> SELECT * FROM Note WHERE Priority='High';
+---------+-------+----------------+----------+-------------+---------+--------+---------------------+---------------------+
| Note_ID | Title | Description    | Priority | Category_ID | User_ID | Status | Created_At          | Updated_At          |
+---------+-------+----------------+----------+-------------+---------+--------+---------------------+---------------------+
|     201 | DBMS  | Study ER model | High     |         101 |       1 | Active | 2026-04-08 21:53:39 | 2026-04-08 21:53:39 |
+---------+-------+----------------+----------+-------------+---------+--------+---------------------+---------------------+
1 row in set (0.00 sec)

mysql> SELECT * FROM Note WHERE Status='Pending';
+---------+-------+-------------+----------+-------------+---------+---------+---------------------+---------------------+
| Note_ID | Title | Description | Priority | Category_ID | User_ID | Status  | Created_At          | Updated_At          |
+---------+-------+-------------+----------+-------------+---------+---------+---------------------+---------------------+
|     202 | CN    | Study DHCP  | Medium   |         102 |       2 | Pending | 2026-04-08 21:53:39 | 2026-04-08 21:53:39 |
+---------+-------+-------------+----------+-------------+---------+---------+---------------------+---------------------+
1 row in set (0.00 sec)

mysql> SELECT U.First_Name, N.Title
   -> FROM User U
   -> JOIN Note N ON U.User_ID = N.User_ID;
+------------+-------+
| First_Name | Title |
+------------+-------+
| Payal      | DBMS  |
| Rahul      | CN    |
+------------+-------+
2 rows in set (0.00 sec)

mysql> SELECT N.Title, C.Category_ID
```

```
   -> FROM Note N
   -> JOIN Category C ON N.Category_ID = C.Category_ID;
+-------+-------------+
| Title | Category_ID |
+-------+-------------+
| DBMS  |       101 |
| CN    |       102 |
+-------+-------------+
2 rows in set (0.00 sec)

mysql> SELECT N.Title, R.Reminder_Time
   -> FROM Note N
   -> JOIN Reminder R ON N.Note_ID = R.Note_ID;
+-------+---------------------+
| Title | Reminder_Time       |
+-------+---------------------+
| DBMS  | 2026-04-10 10:00:00 |
| CN    | 2026-04-11 12:00:00 |
+-------+---------------------+
2 rows in set (0.00 sec)

mysql> SELECT N.Title, A.File_Path
   -> FROM Note N
   -> JOIN Attachment A ON N.Note_ID = A.Note_ID;
+-------+-----------------+
| Title | File_Path       |
+-------+-----------------+
| DBMS  | /files/dbms.pdf |
| CN    | /files/cn.docx  |
+-------+-----------------+
2 rows in set (0.00 sec)

mysql> ALTER TABLE Note ADD COLUMN Extra VARCHAR(50);
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE Note DROP COLUMN Extra;
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE Note MODIFY Title VARCHAR(200);
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE Note RENAME COLUMN Title TO Note_Title;
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> UPDATE Note
   -> SET Status='Completed'
   -> WHERE Note_ID=201;
Query OK, 1 row affected (0.01 sec)
```

```
Rows matched: 1  Changed: 1  Warnings: 0

mysql> DELETE FROM Reminder WHERE Reminder_ID=302;
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> SELECT COUNT(*) FROM Note;
+----------+
| COUNT(*) |
+----------+
|      2      |
+----------+
1 row in set (0.00 sec)

mysql> SELECT COUNT(*) FROM User;
+----------+
| COUNT(*) |
+----------+
|      2      |
+----------+
1 row in set (0.00 sec)

mysql> SELECT MAX(Age), MIN(Age) FROM User;
+----------+----------+
| MAX(Age) | MIN(Age) |
+----------+----------+
|     23 |     21         |
+----------+----------+
1 row in set (0.00 sec)

mysql> SELECT AVG(Age) FROM User;
+----------+
|AVG(Age) |
+----------+
|  22.0000  |
+----------+
1 row in set (0.00 sec)

mysql> CREATE TABLE NOTE_COPY AS SELECT * FROM Note;
Query OK, 2 rows affected (0.03 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM NOTE_COPY;
+---------+------------+----------------+----------+-------------+---------+-----------+---------------------+---------------------+
| Note_ID | Note_Title | DescriptioN| Priority | Category_ID | User_ID | Status| Created_At| Updated_At    |
+---------+------------+----------------+----------+-------------+---------+-----------+---------------------+---------------------+
|201 | DBMS| Study ER model | High | 101 | 1 | Completed | 2026-04-08 21:53:39 | 2026-04-08 21:53:39 |
| 202 | CN | Study DHCP | Medium |102 | 2 | Pending   | 2026-04-08 21:53:39 | 2026-04-08 21:53:39         |
+---------+------------+----------------+----------+-------------+---------+-----------+---------------------+---------------------+
2 rows in set (0.00 sec)

mysql> TRUNCATE TABLE Attachment;
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> DROP TABLE Attachment;
Query OK, 0 rows affected (0.02 sec)

mysql> ALTER TABLE Note COMMENT = 'This table stores notes data';
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

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8. Conclusion

The Notes Management System provides a structured way to store and manage user data, notes, reminders, and attachments. The use of relational database concepts ensures data integrity, efficient retrieval, and scalability, making it suitable for real-world applications.